

DISPOSABLE — CCT P2b workflow thread (living log)

Purpose: One place for **where we are** when Alex drive-bys pull you off the campaign plan.

Companion (science): `_DISPOSABLE_PD25_Alex_board_for_dummies.md` — read that for *what* P2b is; **this file** is *what to do now*.

Protocol: Document-first (Option A). COMPASS updates **YOU ARE HERE** after each substantive turn. Say **anchor** in chat if lost — I'll paste this block.

Delete when: P2b binning settled + Alex briefed + campaign plan triage merged.

YOU ARE HERE (2026-08-24)

Step	Task	Status
A	Articulate back-of-envelope math (below) — enough to answer Alex	✓ In this doc
B	Re-run P2b 4 + 10 tail bins; inspect; adjust if needed	✓ Done — downturn YES (34.0% → 5.6%)
C	Campaign plan triage (OBE vs active)	✓ Below — revisit after B

Drive-by rule: Alex questions **do not automatically supersede** the campaign plan. They get a **fast honest answer** + optional **one** replot. Then return to **YOU ARE HERE** unless Alex explicitly reprioritizes.

One sentence for Alex (binning): “We did the power half of the back-of-envelope — 20 tail bins gave ~10 people per bin; we’re re-running with **10 tail bins** (~20 per bin) so the downturn isn’t over-fit noise.”

A — Back-of-envelope math (minimal)

Question: How many T_j bins can this band support?

Inputs (primary run):

- $N_{\text{band}} = 392$ player-seasons (PPM $z \in [2, 3]$, +DFT, min10)
- Split at within-band **median** $\hat{T}_j \rightarrow n_{\text{above}} \approx N_{\text{band}}/2 \approx 196$

Power (exploratory bar level):

$$\bar{n}_{\text{tail}} = \frac{n_{\text{above}}}{B_{\text{tail}}}$$

Target $\bar{n}_{\text{tail}} \gtrsim 15\text{--}20 \rightarrow$

$$B_{\text{tail}} \lesssim \frac{196}{20} \approx 10$$

Choice:

$$B_{\text{left}} = 4, \quad B_{\text{tail}} = 10 \quad \Rightarrow \quad 14 \text{ bins total}$$

(v0 was $4 + 20 = 24 \rightarrow \bar{n}_{\text{tail}} \approx 196/20 \approx 10$ — too thin for bar-to-bar reads.)

Read rule (not per-bar hypothesis tests):

- Plateau rate** $\bar{p}_{\text{plat}}$ = mean draft rate over coarse / mid bins (pool if needed)
- Tail rate** $\bar{p}_{\text{tail}}$ = mean over **last 2–3** fine bins (pool)
- Downturn** if $\bar{p}_{\text{tail}} < \bar{p}_{\text{plat}}$ (directional; Wilson CI on pooled groups if reporting)

Not done yet: Model cutpoint from K/N , θ , $\gamma \rightarrow$ split on $\hat{T}_j$ (formal memo). v0 uses **median split** as zoom line.

$n \geq 30$ **flag:** Warning for **adjacent-bar** comparison only — not a requirement that every elite bin hit 30. Elite $\hat{T}_j$ is sparse by geography, not because K is tiny inside this band.

Alex’s “~20 bins”: Meant **uniform** binning on X to prune bad grids — **not** “20 fine bins after piecewise.” Piecewise spends bins where congestion lives; **10 tail bins $\approx$ adequate resolution.**

B — Re-run spec

```
python sports/scripts/pass_a_congestion_conditional.py \  
  --plot fixed_ai_tj_knbins --dft --p2b-single \  
  --ai-lo 2.0 --ai-hi 3.0 --min-minutes 10 \  
  --tj-n-low 4 --tj-n-high 10
```

Inspect: downturn still visible? fewer red thin cells? plateau vs tail box sensible?

Adjust knobs if needed: `--tj-n-high 8` (more power) or `12` (more resolution); `--tj-tail-split-q 0.75` (tail-only zoom).

C — Campaign plan triage (2026-08-24)

OBE / moot (do not resume unless Alex re-opens)

Item	Why OBE
P2 heatmap	Parked; Alex PD25 = P2b, not smearing diagnosis
P2 before P2b sequencing	P2b shipped; order obsolete
16-ventile T _j panel	Wrong binning; superseded by P2b
Hunting bin-16 marginal hero dip	Explicit do-not-do
OBPM as canonical	Appendix only — settled
Act II "green-light SCOUT for P1"	Done
Formal K/N → θ cutpoint memo	Open but not blocking Alex drive-by answer

Active now (task at hand)

Item	Owner
P2b bin refinement (4+10 run, inspect)	Charles + COMPASS/SCOUT
Articulate binning to Alex	Charles (use §A + one sentence above)

After task at hand (campaign plan still true)

Item	When
Act III — HAND deck (triptych + P2b + P3 Q4)	Alex: after next week
Paper outline / framing	Same window
§8 Alex read-aloud paragraph	HAND assembly
K/N formal memo	Before claiming model-aligned knee
P4 interval overlap	Optional HAND enrich
PD24 BPM hero rerun	Appendix when bandwidth

Drive-by vs supersede (how to tell)

Alex says...	Treat as...
"Did we do the back-of-envelope?" / "Is 20 bins too many?"	Clarify + maybe one replot — does not cancel Act III or triptych deck
"Stop everything and rewrite the paper"	Supersedes — update campaign plan Status
"Looks good, let's frame the story"	Confirms Act III next — park binning after B satisfied

Thread log

2026-08-24 — Charles: lost; wants A/B/C workflow

- Chose **document-first (A)** for thread tracking.
- This file created. Back-of-envelope math compressed to §A.
- Drive-by protocol: answer Alex first, then return to YOU ARE HERE.

2026-08-24 — B done: 4+10 re-run

- **File:** `basic_data_plots/CCT_draft_rate_fixedAi_Tj_knbins_dft.png` (canonical alias updated)
- **Binning:** 4 coarse + 10 fine (was 4+20)
- **Knee box:** plateau **34.0%** · tail (last 3 bins) **5.6%** · downturn **YES**
- **Charles:** eyeball vs v0 — fewer bars, thicker tail cells; shape holds.

(Entries append below.)